

# SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face & Stanford University

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256M parameters · 0.8 GB RAM · Outperforms 300× larger models

# Large VLMs Are Powerful but Impractical for Edge Deployment

## The Deployment Gap

State-of-the-art VLMs require massive GPU memory:

- Idefics-80B: ~80 GB GPU RAM
- Molmo-A1B-7B: 27.7 GB GPU RAM
- InternVL2-2B: ~4–8 GB GPU RAM

Mobile and edge devices are constrained to 4–8 GB of shared system memory.

**Result: A large gap between research benchmarks and real-world deployability.**

## The SmoIVLM Solution

**256M**

Parameters (smallest model)  
0.8 GB RAM at batch=1

**300×**

Smaller than Idefics-80B  
yet outperforms it

Three model sizes for flexible deployment:

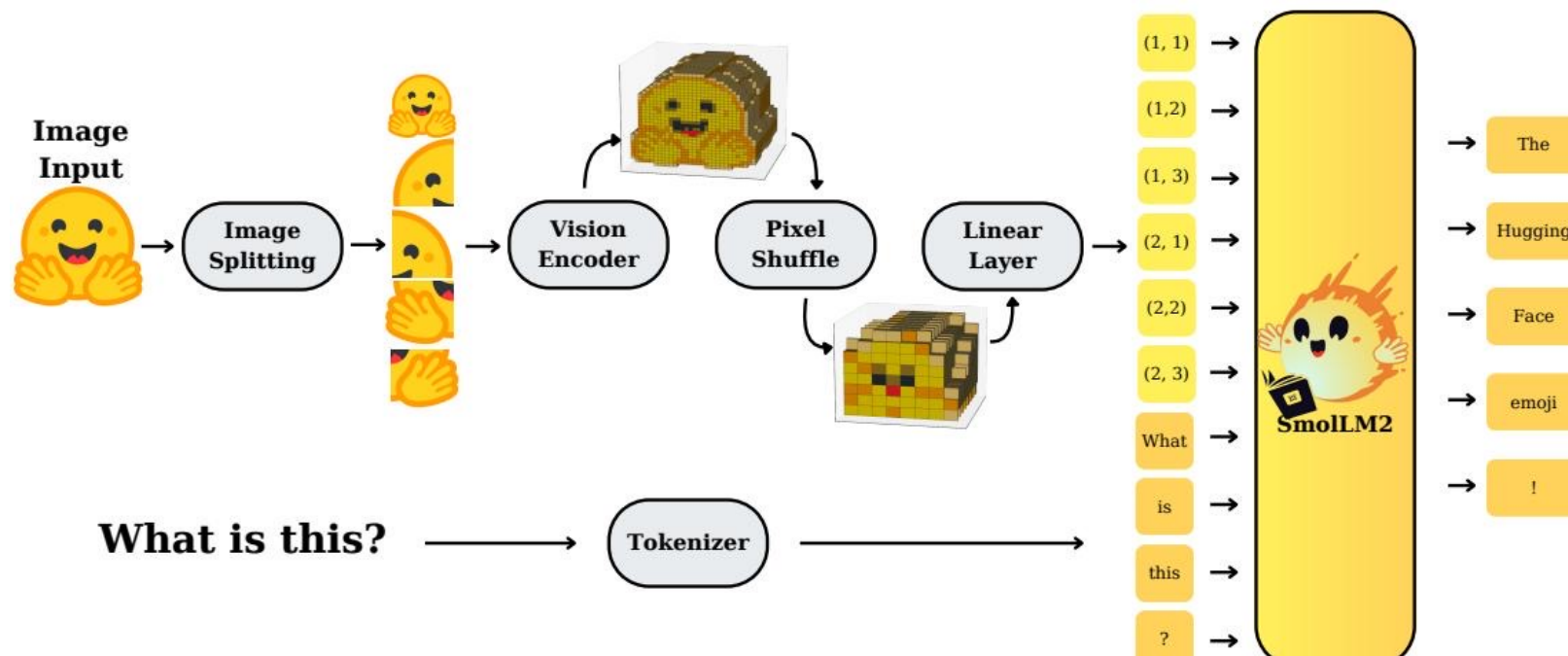
**SmoIVLM-256M** — smartphones, edge devices

**SmoIVLM-500M** — mid-tier laptops, tablets

**SmoIVLM-2.2B** — high-end workstations

**All weights, datasets, code, and the HuggingSnap iOS app are fully open-sourced.**

# SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2



1

## Image Splitting

Input image divided into sub-images for processing

2

## SigLIP Encoder

Vision encoder extracts patch-level features from sub-images

3

## Pixel Shuffle

Space-to-depth compression reduces tokens by  $r^2$  factor

4

## Linear Projection

Maps visual tokens into SmolLM2 input embedding space

5

## SmolLM2 + Text

Concatenated with text tokens for autoregressive generation

## FINDING 1

# Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

## Balanced Encoder-LM Compute

Compact VLMs benefit from smaller vision encoders — a counterintuitive finding.

**93M SigLIP-B/16** + 135M LM  
outperforms

**428M SigLIP-SO400M** + 135M LM

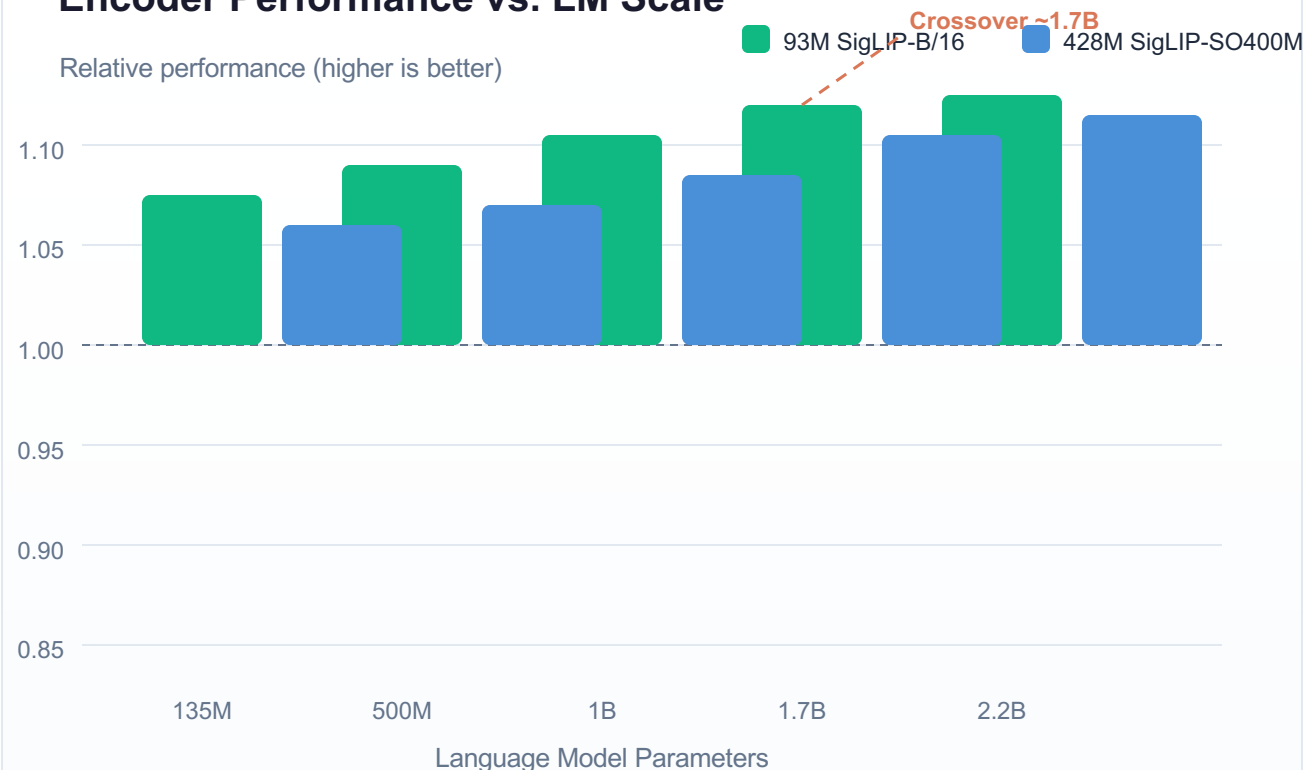
Only at ~1.7B LM scale does the larger encoder justify its 10% parameter overhead.

### Key Insight

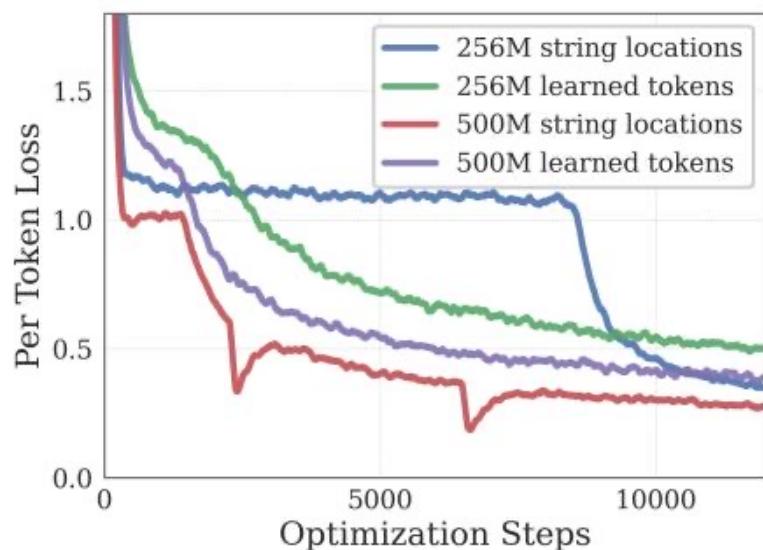
For small LMs, a smaller encoder prevents bottlenecking and improves end-to-end quality.

## Encoder Performance vs. LM Scale

Relative performance (higher is better)



## Aggressive Pixel Shuffle ( $r=4$ ) Cuts Visual Tokens 16x



### Pixel Shuffle (Space-to-Depth)

Rearranges spatial features into depth channels, reducing token count by factor of  $r^2$ .

Source: SmolVLM paper, Figure 6.

### Token Reduction Factor

**4x**

**$r = 2$**

Preferred by larger models (2.2B)  
Moderate compression

**16x**

**$r = 4$**

Preferred by small models (256M/500M)  
Aggressive compression

### Why Aggressive Compression Helps Small Models

- Reduces attention overhead quadratically with token count
- Enables longer context windows within same memory budget
- Small LMs lack capacity to attend over thousands of tokens
- $r=4$  yields better long-context modeling for 256M/500M

**Result: 16x fewer visual tokens → faster inference, lower RAM, better accuracy on small models**

# Extending Context from 2K to 16K Tokens Unlocks Higher Resolution



## RoPE Base Scaling

Position encoding adjustment

- Base increased from 10,000 to 273,000
- Extends effective position encodings 27×
- Enables 16K context without positional aliasing

TECHNIQUE



## Mixed-Context Fine-Tuning

Training data curation strategy

- Mix of long and short sequences in training
- Prevents degradation on short-context tasks
- Maintains fluency across all sequence lengths

TRAINING



## Model-Specific Context

Right-sized for each variant

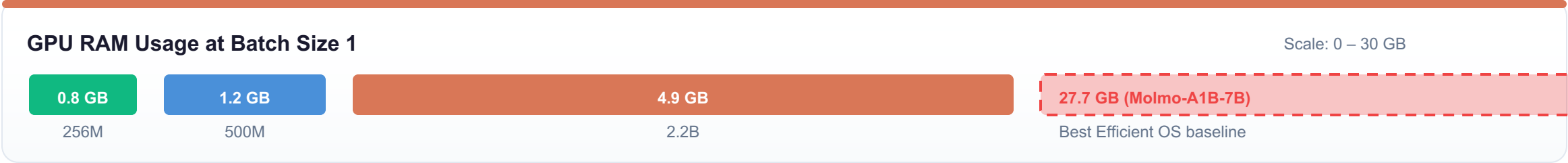
- SmolVLM-2.2B: 16K tokens (highest resolution support)
- SmolVLM-256M: 8K tokens (balanced for edge RAM)
- SmolVLM-500M: 8K tokens (optimal throughput)

DEPLOYMENT

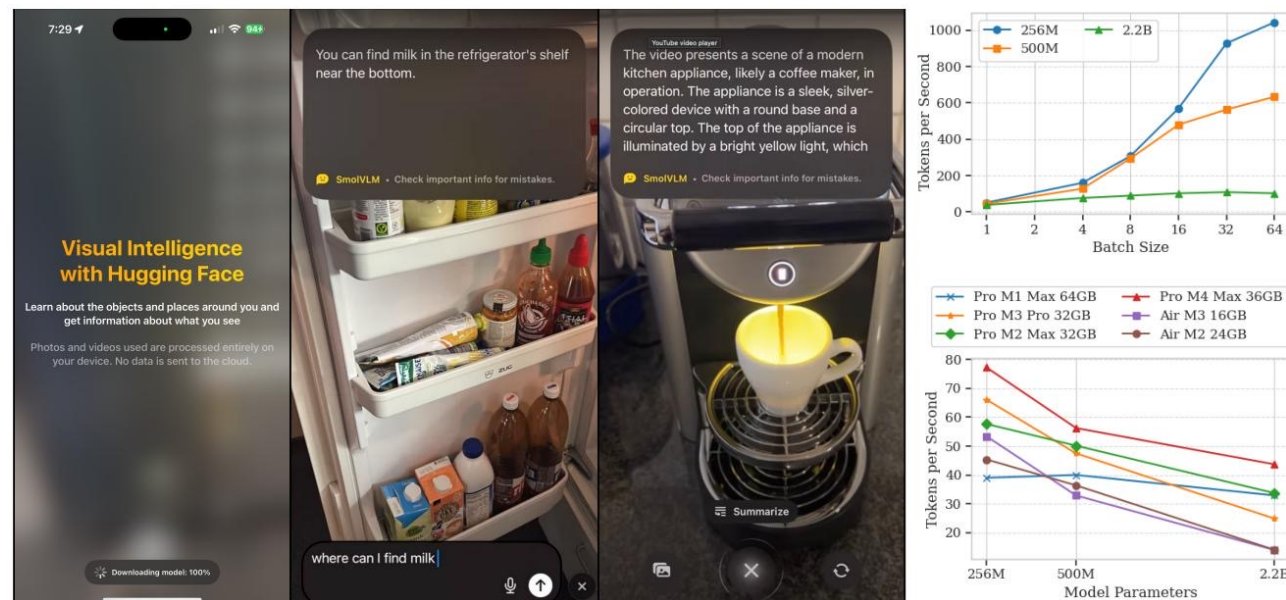
Impact: Performance improves significantly across all benchmarks with context extension from 2K → 16K

# SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

| Capability              | Benchmark | 256M  | 500M  | 2.2B  | Best Efficient OS |
|-------------------------|-----------|-------|-------|-------|-------------------|
| Single-Image            | OCRBench  | 52.6% | 61.0% | 72.9% | 54.7% (Molmo)     |
|                         | AI2D      | 46.4% | 59.2% | 70.0% | 71.0% (Molmo)     |
|                         | ChartQA   | 55.6% | 62.8% | 68.7% | 48.0% (Molmo)     |
|                         | TextVQA   | 50.2% | 60.2% | 73.0% | 61.5% (Molmo)     |
|                         | DocVQA    | 58.3% | 70.5% | 80.0% | 77.7% (Molmo)     |
|                         | ScienceQA | 73.8% | 80.0% | 89.6% | 87.5% (Molmo)     |
| Multi-task              | MMMU      | 29.0% | 33.7% | 42.0% | 33.9% (Molmo)     |
|                         | MathVista | 35.9% | 40.1% | 51.5% | 37.6% (Molmo)     |
|                         | MMStar    | 34.6% | 38.3% | 46.0% | 43.1% (Molmo)     |
| Video                   | Video-MME | 33.7% | 42.2% | 52.1% | 45.0% (InternVL2) |
|                         | MLVU      | 40.6% | 47.3% | 55.2% | 48.2% (InternVL2) |
| Average (13 benchmarks) |           | 44.0% | 51.0% | 59.8% | —                 |



# Real-Time Inference on iPhones: Up to 80 Tokens/Second



**~80**

**tokens/second**

Pro M1 Max 64GB  
SmolVLM-256M



**~1000**

**tokens/second**

Batch size 64  
High-end Apple Silicon



**~55**

**tokens/second**

MacBook Air M3 16GB  
Entry-level Apple Silicon



**100%**

**local inference**

HuggingSnap iOS app  
Zero cloud dependency

## TAKEAWAYS

# Design Principles for Efficient Multimodal Models

### 1 Balance Encoder-LM Compute

Smaller vision encoders (93M) outperform larger ones (428M) for compact LMs.  
Crossover only at ~1.7B LM scale.

### 2 Extend Context Length

RoPE base 10K → 273K unlocks 16K context. Mixed long/short fine-tuning prevents degradation.

### 3 Aggressive Token Compression

Pixel shuffle  $r=4$  gives  $16\times$  reduction for small models;  $r=2$  for large.  
Reduces attention overhead.

### 4 Sub-1GB Inference Is Achievable

SmoVLM-256M runs in 0.8 GB RAM and outperforms Llama-80B.  
Architectural efficiency > raw scale.

### 5 Fully Open-Source Release

Weights, datasets, code, and the HuggingSnap iOS app — all open.  
Promotes reproducible on-device AI.

**"Architectural efficiency matters more than raw scale."**

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